

Fractals

Infinite Detail from Simple Rules

Speaker Scripts

Three speakers · three minutes each · CSE 200 Presentation

How to use this

Each script below is written to be **spoken, not read silently**. The sentences are short on purpose — short sentences are much easier to deliver when you are nervous.

- **[CLICK]** means **advance the slide**. Press the key, *then* keep talking.
- **Bold words** are the ones to lean on. Slow down slightly when you say them.
- Every script is about 390–410 words. At a calm speaking pace (~130 words a minute) that is just about 3 minutes. **Time yourself once out loud** — reading in your head is far faster and will fool you.
- You do not have to memorise this word for word. Learn the **order of the ideas**. If you say them in your own words, that is better, not worse.

The single most useful tip

When you finish a sentence, **stop**. Do not fill the gap with “um” or “so”. A one-second silence feels enormous to you and completely normal to the audience. It is the fastest way to sound like you know what you are doing.

Timing at a glance

Speaker	Covers	Time	Your one job
Speaker 1	History, definition, Sierpinski triangle	3 min	Show that normal measurement <i>breaks</i> .
Speaker 2	Fractal dimension	3 min	Give the broken thing a <i>number</i> .
Speaker 3	Fractals in the real world	3 min	Show where that number <i>shows up</i> .

Names you will have to say out loud

Name	Say it as
Karl Weierstrass	VY-er-shtrass
Georg Cantor	GAY-org CAN-tor
Helge von Koch	HEL-ga fon KOKH
Wacław Sierpiński	VAT-swaf sher-PIN-skee
Felix Hausdorff	HOWS-dorf
Gaston Julia	gas-TOHN ZHOO-lya
Pierre Fatou	pee-AIR fa-TOO
Benoit Mandelbrot	ben-WAH MAN-del-brot
Michael Barnsley	BARNZ-lee
Romanesco	roh-ma-NESS-koh

If you stumble on a name

Say it confidently and move on. Nobody in the room will correct you, and stopping to fix it costs you five seconds and your rhythm.

Speaker 1 — The Surprise

Your goal in one sentence

Convince the audience that ordinary geometry **completely breaks** on these shapes — so that Speaker 2 has a problem worth solving.

Slides: A Crisis in 19th-Century Mathematics → Handoff to Speaker 2 · about 400 words
≈ 3 min

Think about zooming in on a curve. Any curve — a circle, a parabola. Zoom in far enough and it straightens out — it looks like a straight line. For about two hundred years, all of calculus quietly assumed this is **always** true.

[CLICK] In 1872, Karl Weierstrass broke that assumption — on purpose. He built a function that is **continuous everywhere**, but **differentiable nowhere**. It is infinitely jagged — zoom in as far as you like, it never straightens out. No tangent line anywhere.

[CLICK] Mathematicians called these shapes **monsters**. But once the door was open, more came. Cantor built a set of infinitely many points with zero total length. Von Koch drew a curve that is nowhere smooth. Sierpiński made shapes built out of smaller copies of themselves. And Hausdorff gave these shapes a **number** — a fractional dimension, in between the whole numbers we know.

For almost a hundred years, all of this was considered **beautiful but useless**.

[CLICK] Then, in the 1960s, Benoit Mandelbrot — working at IBM — asked a very simple question. **How long is the coast of Britain?**

The honest answer is: it depends on your ruler. Measure with a hundred-kilometre ruler, you get one number. Measure with a one-kilometre ruler, and the coast gets **longer** — every little bay now counts. The length never settles down.

Mandelbrot's insight was that this is not a flaw in the coastline. It means **length is the wrong tool**. And the right tool was exactly that “useless” mathematics. He also had something Cantor and Koch never had — **computers** — so he could finally *see* these shapes. He gave them one name: **fractals**, from the Latin *fractus*, meaning “broken”.

[CLICK] So what is a fractal? Formally: a shape with detailed structure at arbitrarily small scales, whose fractal dimension is greater than its topological dimension. In plain words — **it has detail at every zoom level, and it is built by repeating one simple rule**.

[CLICK] Here is the classic example. Take a triangle. Remove the middle one. Repeat — forever. That is the Sierpiński triangle.

[CLICK] Now watch the numbers. Every step keeps three quarters of the area. So after n

steps the area is $(3/4)^n$ — and that goes to **zero**.

[CLICK] But the perimeter gets multiplied by three-halves at every step. That goes to **infinity**.

[CLICK] So we have a two-dimensional shape with **zero area** and **infinite perimeter**. Our normal tools have completely broken down.

Over to Speaker 2 — if area and perimeter are useless here, what number *do* we use?

Notes for delivery

- You have the most slides of the three. **Do not linger** on the list of names — say them as one quick sweep and move on.
- The area and perimeter slides have the maths already written. Do not read the algebra out loud. Just say the result: shrinks to zero, grows to infinity.
- **If you blank out**, this one sentence carries your whole section: “There are shapes with zero area but infinite perimeter, and normal geometry cannot handle them.”

Speaker 2 — The Number

Your goal in one sentence

Take the thing Speaker 1 broke, and **measure it** — turn “rough” from a word into a number.

Slides: Fractal Dimension section → Handoff to Speaker 3 · about 390 words $\approx$ 3 min

So we are stuck. Area says zero. Perimeter says infinity. Neither one tells us anything useful. We need a different kind of measurement — and the way in is to rethink what **dimension** actually means.

[CLICK] Forget the usual definition for a moment. Let us define dimension by what happens when you **scale** something.

Take a line, and shrink it by half. How many of those copies fit in the original? **Two**.

[CLICK] Take a square, and shrink each side by half. **Four** copies fit.

[CLICK] Take a cube, shrink it by half. **Eight** copies.

Two, four, eight. That is $2^1, 2^2, 2^3$. Look at the exponent — one, two, three. **The exponent is the dimension.**

[CLICK] So in general: if we scale by a factor r , the number of self-similar pieces is $N = (1/r)^D$. And if we solve that for D , we get $D = \log N / \log(1/r)$.

Let us sanity-check it. A line: $\log 2 / \log 2 = 1$. A square: $\log 4 / \log 2 = 2$. A cube: $\log 8 / \log 2 = 3$. It agrees with everything we already believe. Good — the formula is not doing anything strange.

[CLICK] Now let us point it at the Sierpiński triangle. Shrink it by half. How many copies do we get? Not four — **three**. The middle one is missing.

So $D = \log 3 / \log 2$, which is about **1.585**.

That is not a whole number. The Sierpiński triangle is **more than a line, but less than a solid surface**. And that is exactly why area and perimeter failed — we were measuring a 1.585-dimensional object with one-dimensional and two-dimensional tools.

[CLICK] The same method works on the Koch curve. Scale it by one third, and you get four copies. So $D = \log 4 / \log 3$, about **1.262**.

[CLICK] There is one catch. That formula needs **perfect** self-similarity, and a real coastline is not perfectly self-similar. So we use the general version: the **box-counting dimension**. Cover the shape with a grid of boxes of size ε , count how many boxes contain part of the shape, then let ε shrink. The dimension is the limit of $\log N / \log(1/\varepsilon)$.

Same idea — but this one works on anything. Even a photograph.

[CLICK] So here is what we have gained. **Roughness is no longer just a word. It is a number.** We can now say that one coastline is rougher than another, and say *by how much*.

Over to Speaker 3 — where does this number actually show up?

Notes for delivery

- The line-square-cube build-up is the heart of your section. **Go slowly there.** If they follow two, four, eight, they will follow everything after it.
- Say “log three over log two”, not “logarithm of three divided by”. Shorter is clearer out loud.
- You do not need to explain what a logarithm is. Nobody will ask.
- **If you blank out:** “Dimension is just the exponent that tells you how many copies appear when you shrink a shape — and for fractals that exponent is not a whole number.”

Speaker 3 — The World

Your goal in one sentence

Show that this is not a maths curiosity — it is **in nature, in your body, and in the software people use every day.**

Slides: Fractals in the Real World → Conclusion · about 410 words $\approx$ 3 min

We know what a fractal is, and how to measure one. So where is all of this actually hiding?

[CLICK] Everywhere in nature. Romanesco broccoli. A snowflake. An aloe plant. A pine cone. In each one, **a small piece looks like the whole thing.** And on the right is a tree that nobody designed — a computer grew it by repeating one rule.

[CLICK] Here is that idea at its cleanest. Start with a square. Stand a **right triangle** on its top edge, and grow a new square on each leg — so Pythagoras' theorem sits at every joint. Then repeat. At forty-five degrees the two branches match exactly, and you get a symmetric tree. [CLICK] Now tilt that triangle. The branches stop matching — and the tree **leans**, the way a real tree grows toward the light. We changed **one number.**

[CLICK] Now back to Speaker 1's coastline question — with real map data. We walked rulers of different lengths along Britain's coast and plotted it on log-log axes. It comes out a **straight line**, slope about minus 0.28 — a dimension of about **1.28**. Mandelbrot got 1.25. That is Speaker 2's formula, measured off the real world.

[CLICK] And it is inside you, too. Your lungs branch about **twenty-three times.** That packs roughly **seventy square metres** of surface into about five litres of chest. Your blood vessels do the same trick. And doctors now measure the fractal dimension of the vessels in your eye — because when that number drops, it can be an **early sign of disease.**

[CLICK] Now a beautiful application. This fern is drawn by **four rules** — each one shrinks and tilts the whole picture. Six numbers each, plus four probabilities: **twenty-eight numbers in total.** And those twenty-eight numbers redraw the fern at **any resolution.** Zoom in, there is always more leaf. That is **fractal compression** — store the rule, not the pixels.

[CLICK] Games use the same trick. Add smaller and smaller bumps at every scale, and a handful of numbers becomes an endless mountain range. Nobody stored that terrain. It was **grown.**

[CLICK] One honest caveat. **Real fractals stop.** A coastline looks like a coastline until you reach sand grains. Lungs stop at about twenty-three. Only the mathematical ones go on forever. A fractal is a **model** of nature — and every model has a range where it works.

[CLICK] So, to bring it together. Part 1 gave us the surprise. Part 2 gave us the number.

Part 3 gave us the world. Clouds are not spheres, and mountains are not cones — but **roughness has a number, and simple rules make it.**

Thank you.

Notes for delivery

- Your section is the one the audience will enjoy most. **Let them look at the pictures** — pause for a beat after each image appears.
- The lungs number (70 square metres inside 5 litres) always gets a reaction. Say it slowly.
- You close the whole talk. After “Thank you”, **stop and stand still**. Do not trail off or walk away mid-sentence.
- **If you blank out:** “Nature does not store details — it stores a rule and repeats it. That is why fractals are everywhere.”

If someone asks a question

You are not expected to know everything. “**That is a good question, I am not sure**” is a completely acceptable answer and costs you nothing. But these four come up often, so here are short answers.

Is a fractal dimension a real dimension, like length and width?

Not in the everyday sense. It is a measure of **how much detail fills space as you zoom in**. A shape with dimension 1.26 is rougher than a plain line but does not fill an area.

Are real things actually infinite fractals?

No. Real objects are fractal only over a **limited range of scales** — a coastline down to sand grains, lungs for about 23 branchings. Only the mathematical ones are infinite. We covered this on the “Real fractals stop” slide.

Why is the coastline dimension sometimes 1.25 and sometimes 1.28?

Because it depends on the **map data and the range of ruler lengths** you use. Mandelbrot’s 1.25 and our 1.28 come from different datasets. The point is that the number is consistently **above 1** — the coast is genuinely rougher than a line.

Is fractal compression actually used in practice?

It was developed commercially in the 1990s and works beautifully on self-similar images like the fern. For general photographs, JPEG and its successors won out because they are much faster to encode. The **idea** — describe the picture by the rule that makes it — is what matters here.

Before you walk in

Do one full run-through out loud, all three of you, with the slides. Not reading — **speaking**. It will feel awkward once and easy the second time. That single rehearsal is worth more than any amount of re-reading.